# Pregnancy and Plaque Brachytherapy Treatment of Uveal Melanoma A Retrospective Study

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## Abstract

# Background

To examine whether pregnancy affects the prognosis of uveal melanoma (UM) patients undergoing plaque brachytherapy (PBT) and if PBT has any effect on the outcome of such pregnancy.

# Methods

We conducted a single-center retrospective study at the Beijing Tongren Hospital on the population of women with childbearing age who were diagnosed with uveal melanoma and underwent iodine-125 plaque brachytherapy. The outcome of each pregnancy and the status of the fetus was followed-up. Survival analysis were performed using Kaplan-Meier method, with the metastasis and death as endpoints.

# Results

13 patients with 13 full-term pregnancies and 96 non-pregnant women with matched age and tumor size were included. In pregnant group, two patients developed metastasis, one of which died shortly after delivery; local recurrence of UM occurred in 2 patients after or during delivery, and 2 other patients developed secondary glaucoma due to radiation retinopathy. None of the other pregnant patients reported any signs of disease progression. In the control group, 18 metastasis cases including 12 deaths were documented. Pregnant patients and matched control subjects showed no statistical difference in both Metastasis-free survival (hazard ratio (HR): 0.66, 95% confidence interval (CI): 0.15–2.86; $P = 0.576$) and overall survival (HR: 0.48, 95% CI: 0.06–3.66; $P = 0.464$). All pregnant patients carried the pregnancy to term and delivered healthy babies with no report of placental or infant metastases to date.

# Conclusion

Pregnancy exerted no adverse effects on the prognosis of UM patients who receive PBT. While PBT had no significant effect on maternal fertility, and did not show teratogenic effect on the fetus so far, long-term effects require further follow-up studies.

## Introduction

Uveal melanoma (UM) is the most common primary intraocular malignant tumor in adults(1), with a 5-year survival rates ranging from 25–66%(2). Most cases may result in poor vision or even enucleation(3). Plaque radiotherapy is a treatment that sutures a curvilinear radioactive plaque onto the sclera, precisely over the tumor, to deliver trans-scleral radiation to the UM. In expert hands and with careful patient selection, plaque radiotherapy can achieve comparable overall life prognosis to other management techniques such as enucleation(4, 5) with globe salvage in 95% of patients, often with useful vision retention(3). As a result, this treatment has been used as the first line therapy for most UM patients in our hospital.

According to multiple reports, melanoma is more commonly seen in the elder, male subjects(6, 7). However, in clinical practice, a significant number of women of childbearing age (15–45 years old) still face the threat of UM. Also despite the fact that cancer survivors were 27% less likely to give birth than the general population, survivors of non-reproductive organ cancer had practically similar relative birth rates as the background population(8). Therefore, in treating these patients, it is important to take care of their reproduction needs and concerns.

There has been conflicting data and opinions on how the physiological process of pregnancy affects UM on the prognosis. Given that pregnancy induces physiological changes in the eye(9), it can be assumed that pregnancy may have an effect on intraocular tumors such as UM. Hartage et.al. found that history of pregnancy had an increased risk of UM(10, 11) which suggested that female hormones may have some effect on the pathophysiological process of UM. While other authors reported exact opposite findings that a history of pregnancy decreased risk of developing UM(12). For prognosis, multiple researchers found no appreciable influence on that of UM at all(13, 14) while Egan et.al. believes a history of childbearing affords protection from metastatic death in intraocular melanoma(15). These various findings make the effect of pregnancy on UM an interesting and worthy aspect of study.

As for the effect of UM and its treatment on pregnancy, since plaque brachytherapy is a treatment of radiation, though localized and minimal, its impact on maternal fertility and the fetus has become a concern for many patients and clinicians.

This study aimed to evaluate the prognosis of women of childbearing age who received plaque brachytherapy in our hospital, and the outcomes of their pregnancy.

## Materials and Methods

A single-center retrospective study was performed on the medical records of clinically diagnosed UM patients who were of childbearing age and treated with iodine-125 plaque brachytherapy by one physician (Dr. Yue-Ming Liu) at Beijing Tongren Hospital, Capital Medical University between January 1, 2012 and December 31, 2019. This study was approved by the ethical committee of Beijing Tongren Hospital of Capital Medical University. Research adhered to the tenets of the Declaration of Helsinki. All adult subjects (Age ≥ 18 years old) involved in the clinical data used in this study signed written informed consent. The informed consent of subjects who were less than 18 years old was signed by their guardian.

The patients were diagnosed with UM using ophthalmoscopy, ultrasonography, and fundus fluorescein angiography, and were all immediately treated by plaque brachytherapy (PBT). The size of the plaques was designed by the nuclear medicine department of Beijing Tongren hospital according to the maximum tumor base diameter of each patient. Then the device was placed precisely on the sclera over the tumor via surgery. The prescription dose at the top of the tumor was 100Gy, and the time to reach the prescription dose could be calculated according to the activity of the plaque and the height of the tumor. When the prescription dose was reached, the device was removed. All patients were evaluated at regular intervals after PBT treatment and were followed up until Oct. 2021. Since none of the patients gave birth at Tongren hospital, the outcome of each pregnancy and the status of the fetus was followed-up through physicians of birth hospital.

The clinical data of 282 consecutive childbearing-age female patient including age, gender, date of treatment was collected. Tumor diameter was measured by B-scan ultrasonography. For pregnant patients, the date of delivery, pre-pregnancy and post-pregnancy clinical features of the tumor and disease related events, if any, were carefully documented. Since all pregnancies were carried to term, date of conception was estimated 40weeks before delivery. A total of 13 women had 13 full-term pregnancies after the diagnosis and PBT treatment of uveal melanoma. Since age is an independent prognostic factor for metastases of UM, and number of subjects did not permit formal multivariate analyses that would adjust for this age discrepancy, we selected non-pregnant women who were of comparable age and who had similar tumor base diameter and tumor height as control subjects. Selections were made without knowledge of the survival status of the patient.

We used a commercially available statistical analysis program (SPSS 25.0; IBM-SPSS Inc., Chicago, IL, USA) for the statistical analysis of the data. The statistical significance of differences between the pregnant group and the control group was calculated using independent-sample t test and chi-square test. Analysis-survival comparisons were based on the Kaplan-Meier method, with the endpoint being diagnosis of metastases and death. There were no deaths due to other causes. Comparability of survival distributions was evaluated using the log-rank test.

## Results

# Demographics of patients

Ultimately, a total of 109 patients were included in our study with a median age of 32 years old (range: 15–38 years old), including 13 patients who had pregnancy after PBT treatment of UM (pregnant group, median age 30, range 24–38) and 96 patients who met our criteria as control group (median age 33 range 15–37). For all included patients, the height of the tumors was between 2.4 mm and 14.6 mm (mean: 7.1 ± 2.2). The mean tumor basal diameter was 13.0 ± 2.9 mm (5.7 mm-20.7 mm) (Table 1). There was no difference in onset age (median, 33 and 30 years old, respectively; $P = 0.391$), tumor base diameter (mean, 13.1 ± 2.8 and 12.8 ± 3.5 mm, respectively; $P = 0.176$), or tumor height (mean, 7.1 ± 2.2 and 7.4 ± 2.9 mm, respectively; $P = 0.416$) between the two groups. The clinical features and pregnancy outcomes of pregnant patients were shown in Table 2.

Table 1
Basic characteristics of included patients.

| | **Pregnant group (n = 13)** | **Non-pregnant group (n = 96)** | ***P* value** |
|---|---|---|---|
| Mean Age | 30 | 31 | 0.391 |
| Max | 38 | 38 | |
| Min | 24 | 15 | |
| Tumor Base | 12.8 ± 3.5 | 13.1 ± 2.8 | 0.176 |
| Tumor Height | 7.4 ± 2.9 | 7.1 ± 2.2 | 0.416 |
| Metastasis case (%) | 2(15.38%) | 18(18.75%) | 0.769 |
| Death case(%) | 1(7.69%) | 12(12.5%) | 0.616 |

Table 2
Clinical features and pregnancy outcomes of pregnant patients after PBT (n = 13).

| Case No. | Age at Diagnosis | Clinical Features in Primary Survey | | Time of PBT | Pre-Pregnancy Clinical Features | | Date of Dilivery | Post-Pregnancy Clinical Features | | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Base | height | | Base | height | | Base | height | |
| 1 | 30 | 17.0 | 10.0 | 2016/10/10 | 15.8 | 5.6 | 2020/2/1 | 15.8 | 4.2 | Alive |
| 2 | 28 | 11.3 | 5.0 | 2014/9/22 | 8.4 | 3.3 | 2019/9/1 | NA | | Alive |
| 3 | 37 | 11.3 | 8.9 | 2015/4/20 | 13.0 | 7.5 | 2018/3/1 | 12.9 | 6.6 | Discovered metastasis on 2018/1/1, Dead on 2018/4/1 |
| 4 | 30 | 19.4 | 4.4 | 2019/3/11 | 12.0 | 3.6 | 2020/8/1 | NA | | Enucleation during pregnancy |
| 5 | 25 | 11.4 | 8.1 | 2012/3/8 | 12.0 | 5.7 | 2016/12/1 | 11.6 | 3.7 | Alive |
| 6 | 31 | 10.6 | 6.0 | 2015/2/2 | 8.4 | 6.1 | 2016/4/1 | 9.3 | 3.8 | Alive |
| 7 | 26 | 11.3 | 7.7 | 2013/11/27 | NA | | 2020/9/1 | NA | | Enucleation in 2014 |
| 8 | 34 | 17.0 | 14.6 | 2016/6/6 | NA | | 2020/5/1 | NA | | Enucleation in 2017/2 |
| 9 | 32 | 12.4 | 6.4 | 2015/3/23 | 9.0 | 3.9 | 2016/10/1 | 8.4 | 3.2 | Alive |
| 10 | 38 | 7.5 | 4.9 | 2015/5/18 | 7.1 | 4.2 | 2018/4/1 | 7.3 | 2.8 | Alive |
| 11 | 25 | 14.7 | 7.5 | 2018/3/19 | 11.9 | 5.5 | 2020/10/1 | 11.2 | 4.8 | Alive |
| 12 | 26 | 14.5 | 8.6 | 2013/11/6 | 20.7 | 8.6 | 2017/12/1 | NA | | Enucleation after delivery(tumor body filled vitreous chamber)<br><br>Discovered metastasis on 2018/1/31 |
| 13 | 24 | 8.1 | 4.1 | 2013/3/1 | 6.4 | 0.9 | 2017/4/1 | NA | | Alive |

# Patient outcome

The mean follow-up time was 67.0 ± 27.7 months (median:66.0 months, range:21.0 to 116.0 months). In 20 patients, metastasis had developed by the close of the study; metastatic disease was diagnosed in one woman 2 months after the birth of a child, 50 months after treatment; metastasis developed in one woman during pregnancy, and the patient died 1 month after delivery; in 18 women, with no pregnancies after diagnosis, metastases developed between 5 months and 6.8 years (median, 3.1 years) after treatment, unfortunately, twelve of which died.

In the pregnant group, local recurrence of choroidal melanoma occurred in 2 patients after or during delivery, and 2 other patients developed secondary glaucoma due to radiation retinopathy. These 4 patients were treated by enucleation immediately or after delivery. Enucleation and pathology examination were conducted at local hospital or our own. All pathology result confirmed the diagnosis of uveal melanoma and revealed necrosis of most of the tumor tissue. None of the other pregnant patients who had examination before and after pregnancy (n = 7) showed rapid tumor growth or other signs of disease progression. Local tumor recurrence, enucleation and pathological data of the control group were absent from the research.

Surviving women in the study were followed an average of 5.3 years after diagnosis (range, 1.8 years to 9.7 years) and the follow-up pregnancy ranged were from 7 months to 11.1 years (mean, 4.6 years). Thirteen deaths were documented and were all caused by metastatic melanoma. Metastasis and death rate of the two groups pregnant / non-pregnant were 15.38% / 18.75% and 7.69% / 12.5%, respectively. Results among the two groups (pregnant v.s. non-pregnant) showed no statistical significance of metastasis ($P = 0.77$) and death ($P = 0.62$).

Given the small sample size of patients who got pregnant, metastasis-free survival rate and survival rate may not reflect true characteristics of the two groups. We compared Metastasis-free survival months and overall survival months of both groups using Kaplan-Meier method (Fig. 1). Pregnant patients and matched control subjects showed no statistical difference in both aspects (hazard ratio (HR): 0.66, 95% confidence interval (CI): 0.15–2.86, $P = 0.576$ and HR: 0.48, 95% CI: 0.06–3.66; $P = 0.464$).

# Pregnancy

All 13 patients in the pregnant group began their pregnancy some time (median: 31 months, range: 3 to 71 months) after PBT treatment. All pregnancies were conceived naturally and were carried to term without any report of adverse events. All pregnant patients delivered healthy babies and with no report of placenta or infant metastases.

## Discussion

This is the first study to focus on the impact of pregnancy on the prognosis of UM patients treated with PBT and that based on Chinese population. Also, we were the first to discuss weather UM or PBT treatment has any effect on the physiology process of pregnancy and its outcome.

Above all, our results show that the survival and metastasis rates of these patients is similar to that of women of childbearing age who had no pregnancy after treatment. This result is consistent with the absence of estrogens and progestogens in UMs(11, 16–18). The only two previous clinical studies(13, 14) which focused on pregnancy occurred after the treatment of UM also supports our findings. Similar to our report, both studies treated most their patients with conservative treatment primarily such as proton beam irradiation or brachytherapy. Therefore, These findings of similar prognosis among patients who had or didn’t have children after PBT may also be the result of the cessation of mitotic activity and tumor cell necrosis in most patients after local radiotherapy(19).

However, there have been occasional cases of significantly increased UM during pregnancy over the past few decades(16, 17). Lee et.al reported a 31-year-old woman with a 29-week pregnancy was diagnosed with uveal melanoma(17). She refused any treatment for the remaining three months of her pregnancy and the tumor increased in size from 8×5 $mm^2$ to 13×12 $mm^2$. Recently, Yue et.al. reported the first Chinese case of accelerate growth of UM during pregnancy and the tumor showed a rapid growth rate and extended to the orbit with muscle invasion. Similar to the previous case, the patient did not receive any treatment before the end of pregnancy, also there was no detailed medical report but only the subjective feelings of the patient prior to pregnancy. Combining previous retrospective case-control studies and reports of these two cases, we suggest that UM patients treated with local radiation therapy is probably unaffected by hormonal and ocular physiological changes during pregnancy.

Also, similar to some previous studies, we demonstrated that UM had no significant impact on maternal fertility for there had been hundreds of cases reported of UM discovered during pregnant and UM patients getting pregnant(11, 14, 16, 20–22). Among all these reports, including our own, and the vast number of cases that they have reported, we found only one case report of placental metastasis attributed to choroidal melanoma(23), and none fetal metastasis. Therefore, we can draw a preliminary conclusion that it is extremely unlikely for choroidal melanoma to metastasize to fetus. Additionally, we also proved that the plaque brachytherapy with localized radiation also has neither significant effect on maternal fertility, nor teratogenic effect on the fetus.

Some limitations of this study should be also noted. First of all, like previous studies, the number of pregnant cases in this study was limited. UM is a rare disease with most patients presenting between the ages of 50 years and 70 years(1). It is estimated to have an incidence rate of four to eight per million, with a reported higher rate among men(2, 24). Therefore, few women of childbearing age have a history of uveal melanoma. Additionally, although survivors of non-reproductive organ cancer had similar birth rates as the background population(8), in clinical practice, many patients gave up the idea of having children upon diagnosis of malignant tumor. These factors have limited the number of cases in studies of pregnancy in UM patients after treatment. Secondly, according to follow-up, gross examination of the placenta did not show any evidence of metastatic disease, and since none of the pregnant patients gave birth in our hospital, none of the specimen underwent microscopic examination. We believe that pathological examination of the placenta should be performed as described by previous study(23) to accurately exclude placental metastases.

In the past 20 years, few articles have focused on this issue, and we hope to attract researchers' attention to women of reproductive age with malignant tumors. Further clinical and basic studies are needed to draw more definitive conclusions.

# Conclusions

Despite these controversies and limitations, the results of this study are a great piece of good news for UM patients with fertility needs, and also provide valuable reference for clinicians to provide fertility guidance to UM patients. We demonstrated that the prognosis of women with uveal melanoma who underwent PBT is not influenced by pregnancy. Also, plaque brachytherapy had no significant effect on maternal fertility, and did not show teratogenic effect on the fetus so far. Therefore, if a patient conceives after the diagnosis of UM and underwent PBT, pregnancy termination is not necessary, but a close ophthalmological follow-up until delivery and a regular ophthalmologic examination of the baby is strongly advised.

# Declarations

### Ethics approval and consent to participate

This study was approved by the ethical committee of Beijing Tongren Hospital of Capital Medical University. Research adhered to the tenets of the Declaration of Helsinki. All adult subjects (Age≥18 years old) involved in the clinical data used in this study signed written informed consent. The informed consent of subjects who were less than 18 years old was signed by their guardian.

### Consent for publication

Not applicable

### Availability of data and materials

Research data of this study are not publicly available on legal grounds. Data can be made available from the corresponding author on a reasonable request.

### Declaration of Interests

All authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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### Authors' contributions

Conception and design of the research: HTW, YML and WBW; Acquisition and interpretation of the data: HTW, YML, WBW and DL, RHZ, WDZ, HYL; Statistical analysis and writing of the manuscript: HTW, YML, DL, RHZ and HYL; Critical revision of the manuscript: HTW and WBW.

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## Figures

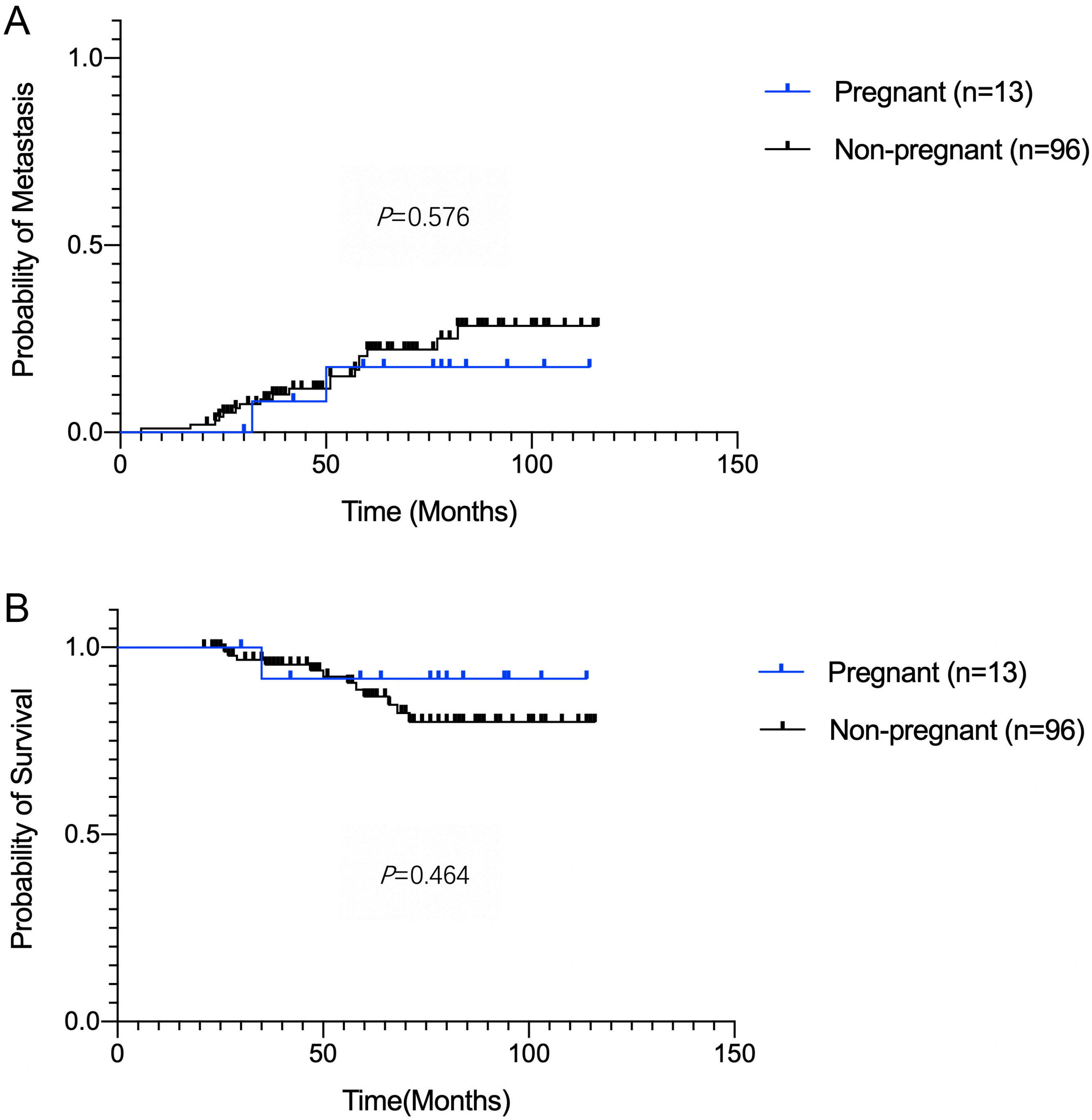


**Figure 1**

Kaplan-Meier survival curves for the metastasis (A) and overall survival (B) in the two groups.